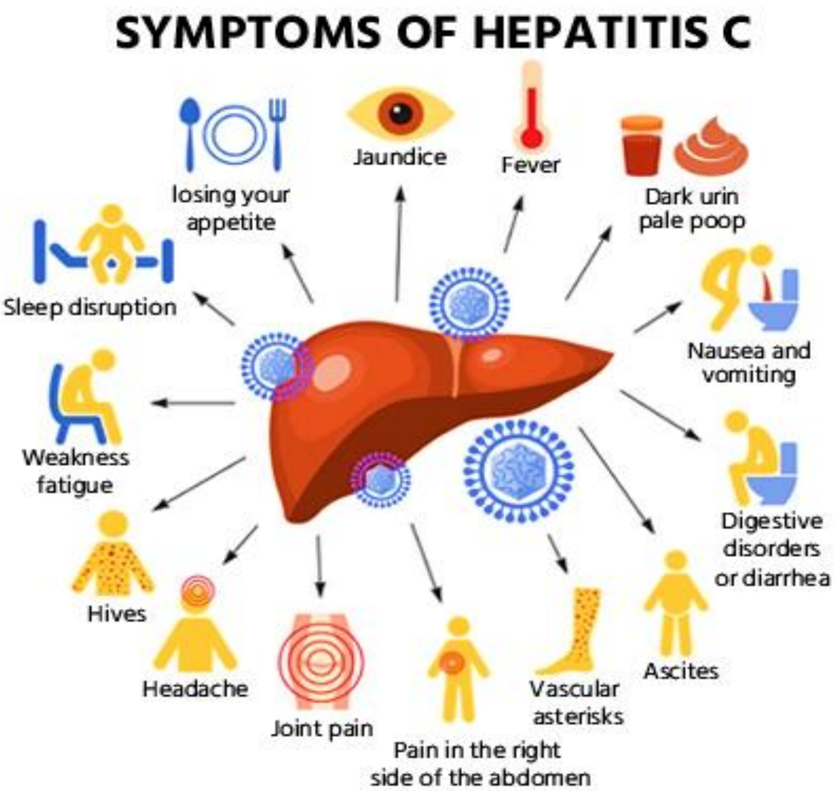
Cairo University
Faculty of Economics and Political Science
Statistics department
4rd level / Arabic section
introduction to Data Science

Hepatitis C Virus (HCV) for Egyptian patients



PRESENTED BY

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SUPERVISION

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Introduction:

Chronic Hepatitis C Virus (HCV) is a major public health issue particularly in Egypt that impacts millions globally, presenting a significant risk for liver-related diseases and deaths. One of the most important aspects of HCV is the long-term impact on liver health, which can lead to conditions like fibrosis, cirrhosis, and, in severe cases, liver cancer. Baseline histological staging quantifies liver fibrosis extent via biopsy and is a critical clinical indicator in managing HCV patients.

Understanding the factors that influence HCV helps in identifying patients at higher risk of disease progression, enabling timely and targeted interventions, and studying this variable can aid in predicting the response to antiviral therapy and the potential for reversing liver fibrosis with successful treatment.

This dataset includes detailed information about patient health conditions, laboratory test results, and treatment regimens, offering opportunities for predictive modeling, data analysis, and public health studies. Its findings can contribute to the understanding of HCV treatment effectiveness and guide future healthcare policies in Egypt and beyond.
The treatment of Hepatitis C involves a diverse range of patients with varying characteristics, comorbidities, and responses to therapy.

we aim to predict whether the patient still has c virus after the end of treatment or not.

Data description:

The Hepatitis C Virus (HCV) dataset is downloaded from the **UC Irvine Machine Learning Repository**. (https://archive.ics.uci.edu/dataset/503/hepatitis+c+virus+hcv+for+egyptian+patients) It contains demographic, clinical, and laboratory information collected from patients who underwent treatment dosages for HCV about 18 months in Egypt, as it is one of the highest countries with prevalence rates of HCV globally.

The dataset consists of 1,384 observations and 29 variables but we will take 20 variables only to reduce the variables that we will study to the important variables and to enhance the efficiency of the analysis.

We aim to perform & apply logistic regression model to predict whether the patient still has **c virus** after the end of treatment or not, as well as applying machine learning method as: Decision tree & KNN, So, we will take 11 numerical variables, and 9 categorical variables beside the response variable, which are:

– Response variable:

HCV: represents if the patient has virus c after the end of treatment or not. Categories:

- Absent: 1
- Present: 2

– Numerical Variables:

1. **Age:** representing the patient's age in years.
2. **BMI (Body Mass Index):** A measure of body fat based on height and weight. It’s commonly used to assess whether an individual is underweighted, normal weight, overweight, or obese.
3. **WBC (White Blood Cell Count):** The count of white blood cells in a given volume of blood, typically used to assess the immune system’s status. An elevated WBC count may suggest infection or inflammation.
4. **RBC (Red Blood Cell Count):** The number of red blood cells in a given volume of blood. RBCs are responsible for oxygen transport in the body. Abnormal levels can indicate anemia or blood-related issues.
5. **HGB (Hemoglobin):** The protein in red blood cells that carries oxygen. Low hemoglobin can indicate anemia or other blood disorders.
6. **Plat (Platelet Count):** The count of platelets in the blood, which are involved in clotting. High or low platelet counts can be indicative of bleeding disorders or bone marrow issues.

7. **AST.1 (Aspartate Aminotransferase 1):** AST is an enzyme found in the liver, heart, muscles, and other tissues. Elevated levels may indicate liver damage or disease.
8. **ALT.1 (Alanine Aminotransferase 1):** It is another liver enzyme, specific to the liver. Like AST, elevated ALT levels can signal liver damage, often used together to assess liver health.
9. **ALT.48 (Alanine Aminotransferase at 48 Weeks):** ALT levels measured 48 weeks into treatment or monitoring. The variable represents how the enzyme level changes over time in response to treatment.
10. **RNA.Base (Base Level of RNA):** The RNA level before starting treatment (baseline). This is likely a measure of viral load in a disease like Hepatitis C, which has a high correlation with the severity of the infection.
11. **RNA.EOT (RNA Level at End of Treatment):** RNA level measured at the end of the treatment. A significant reduction in RNA at the end of treatment is often used as a marker for successful treatment.

Categorical Variables: -

1. **Gender:** indicating the patient's gender, with categories:
   - Male: 1
   - Female: 2
2. **Fever:** indicates whether the patient has experienced a raised body temperature, with categories:
   - Absent: 1
   - Present: 2
3. **Nausea/Vomting:** represents whether the patient has experienced nausea and/or vomiting, with categories:
   - Absent: 1
   - Present: 2
4. **Headache:** indicates whether the patient has experienced headaches, with categories:
   - Absent: 1
   - Present: 2
5. **Diarrhea:** indicates whether the patient has experienced diarrhea, with categories:
   - Absent: 1
   - Present: 2
6. **Fatigue & Generalized Bone Ache:** indicates whether the patient has experienced fatigue and/or bone aches, with categories:
   - Absent: 1
   - Present: 2
7. J**aundice:** refers to the yellowing of the skin and eyes due to elevated bilirubin levels in the blood, resulting from liver dysfunction, with categories:
   - Absent: 1
   - Present: 2
8. **Epigastric Pain:** refers to pain or discomfort in the upper central region of the abdomen, directly above the stomach, with categories:
   - Absent: 1
   - Present: 2
9. **Baseline Histological Staging:** represents the degree of liver fibrosis at the start of treatment, assessed through liver biopsy. Fibrosis is the accumulation of scar tissue in the liver due to chronic inflammation, this variable is often scored on a scale from **1** to **4**, where higher values indicate more severe liver damage. with categories:
   - Mild Fibrosis: 1
   - Moderate Fibrosis: 2
   - Severe Fibrosis: 3
   - Cirrhosis: 4

Cleaning & preparing the data:

First, we convert variable types to the correct types, then we check if we have any missing in the data and we made sure that there are no missing values in our data. There are also no duplicates.

- To check the outliers, we use boxplot:

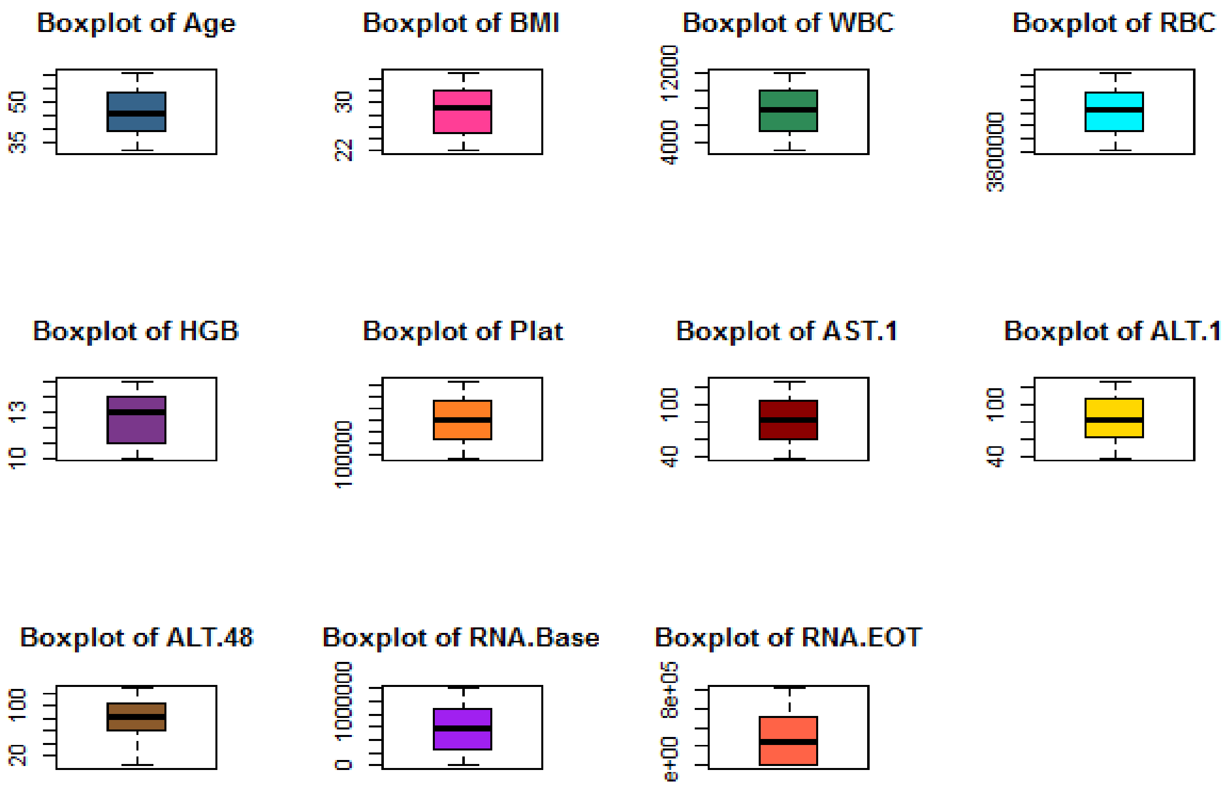

Figure 1: Boxplot for numerical variables

Based on the concept of boxplot, Outliers are points that fall outside the whiskers and shown as dots or stars beyond the whiskers.

From this output we can conclude that we don't have any outlier in any variable, so we don't have to do the process of treat outliers.

- ### Descriptive statistics:
  - #### Descriptive measures for numerical variables:

Table 1: summary of numerical variables

	Age	BMI	WBC	RBC	HGB	Plat	AST.1	ALT.1	ALT.48	RNA.Base	RNA.EOT
Mean	46.1	28.6	7501.0	4429365.6	12.5	159125.5	84.1	84.7	83.2	585737.8	288950.2
Sd	8.6	4.03	2661.3	347619.7	1.7	39000.0	25.7	25.7	26.0	350929.6	265271.3
Median	46.	29.0	7492.0	4442100.0	13.0	159008.5	85.0	84.0	83.0	588841.5	255961.5
Trimmed	46.0	28.6	7502.0	4432505.4	12.6	159008.4	84.2	84.8	83.2	583549.1	267229.0
Mad	11.8	5.9	3503.3	448203.3	2.9	49573.7	34.1	32.6	32.6	452088.4	379481.1
Min	32	22	2991	3816422	10	93013	39	39	5	11	5
Max	61	35	12101	5018451	15	226428	128	128	128	1200762	808142
Range	29	13	9110	1202029	5	133415	89	89	123	1200751	808137
Skew	0.05	-0.06	0.01	-0.06	-0.04	0.01	-0.01	-0.02	-0.04	0.02	0.40
Kurtosis	-1.22	-1.22	-1.23	-1.19	-1.30	-1.22	-1.20	-1.17	-1.09	-1.19	-1.21
Se	0.28	0.13	85.54	11172.92	0.06	1253.51	0.83	0.83	0.84	11279.31	8526.14

The Age variable ranges from 32 to 61 years, with average age is approximately 46 years, indicating a middle-aged population. BMI values range from 22 to 35, with a mean of 28.65, suggesting many patients are overweight or obese. WBC counts vary from 2991 to 12101, with a mean of approximately 7501, indicating that most participants likely have no active infection at the time of measurement. RBC counts range from 3816422 to 5018451, with a mean of approximately 4429366, indicating normal red blood cell counts. HGB levels range from 10 to 15, with a mean of approximately 13, showing normal hemoglobin levels for most patients. Platelet counts vary from 93013 to 226428, with a mean of 159125, suggesting normal blood clotting ability in most participants. Liver enzyme levels (AST and ALT) are elevated, with AST.1 ranging from 39 to 128 and a mean approximately 84, while ALT.1 ranges from 39 to 128 with a mean of approximately 85, suggesting liver stress or damage and ongoing liver involvement despite treatment. RNA viral loads show significant variation, with baseline levels (RNA.Base) ranging from 11 to 1200762 and a mean of 585737, suggesting varied infection severity, with reductions in viral load during treatment. The RNA.EOT ranging from 5 to 808142, with mean 288950, indicating a partial response to therapy and a reduction in viral load.

Most of the variables is symmetric but the variable RNA.EOT has slight positive skew.

Variables like WBC and RNA levels have high SDs, showing substantial variation among participants, while BMI and HGB have lower SDs, reflecting more consistency. SE is small for all variables due to the large sample size, ensuring reliable mean estimates despite the variability.

- **Variance covariance matrix:**

```
                   Age           BMI           WBC           RBC
Age       7.467886e+01 -1.423716e+00  8.005320e+02 -2.062395e+04
BMI      -1.423716e+00  1.624403e+01  2.769405e+02  8.627557e+03
WBC       8.005320e+02  2.769405e+02  7.082932e+06 -7.974964e+06
RBC      -2.062395e+04  8.627557e+03 -7.974964e+06  1.208395e+11
HGB      -7.324348e-01  5.856573e-01 -1.938893e+01  7.166200e+03
Plat      1.728237e+03 -3.527726e+02 -1.734186e+06  9.488659e+07
AST.1    -6.417813e+00  6.300734e-01  3.991480e+02 -6.979889e+04
ALT.1     2.058787e-01  4.217897e+00 -5.627690e+03 -9.853983e+04
ALT.48    7.123926e+00 -3.943623e+00 -4.433673e+02 -7.980227e+05
RNA.Base  3.248148e+04 -3.910006e+04  2.076172e+07 -2.546455e+09
RNA.EOT  -1.057965e+05 -1.410159e+03 -6.812457e+06 -3.342074e+06
                   HGB          Plat         AST.1         ALT.1
Age      -7.324348e-01  1.728237e+03 -6.417813e+00  2.058787e-01
BMI       5.856573e-01 -3.527726e+02  6.300734e-01  4.217897e+00
WBC      -1.938893e+01 -1.734186e+06  3.991480e+02 -5.627690e+03
RBC       7.166200e+03  9.488659e+07 -6.979889e+04 -9.853983e+04
HGB       2.972814e+00 -2.374575e+03 -7.161281e-01 -4.822190e-01
Plat     -2.374575e+03  1.521007e+09 -3.896157e+04  4.909445e+04
AST.1    -7.161281e-01 -3.896157e+04  6.647431e+02  3.787300e+01
ALT.1    -4.822190e-01  4.909445e+04  3.787300e+01  6.650129e+02
ALT.48   -1.183314e+00 -2.250325e+04  2.591495e+00  3.078421e+01
RNA.Base -2.690102e+04 -2.670470e+08  4.825833e+04  2.715120e+05
RNA.EOT   9.419927e+03  6.861157e+08 -4.819738e+05 -1.251111e+05
                ALT.48      RNA.Base       RNA.EOT
Age       7.123926e+00  3.248148e+04 -1.057965e+05
BMI      -3.943623e+00 -3.910006e+04 -1.410159e+03
WBC      -4.433673e+02  2.076172e+07 -6.812457e+06
RBC      -7.980227e+05 -2.546455e+09 -3.342074e+06
HGB      -1.183314e+00 -2.690102e+04  9.419927e+03
Plat     -2.250325e+04 -2.670470e+08  6.861157e+08
AST.1     2.591495e+00  4.825833e+04 -4.819738e+05
ALT.1     3.078421e+01  2.715120e+05 -1.251111e+05
ALT.48    6.785533e+02  3.198220e+05 -6.350259e+04
RNA.Base  3.198220e+05  1.231516e+11  1.136345e+09
RNA.EOT  -6.350259e+04  1.136345e+09  7.036889e+10
```

Matrix_1: Variance covariance matrix

This matrix provides insight into the variability of each variable and their linear relationships with one another. The diagonal elements represent the variances of individual variables. The off-diagonal elements indicate the covariance between different pairs of variables. As Age has a variance of approximately 74.68, indicating moderate variability in age. BMI has a variance of 16.24, showing that BMI is less variable compared to age. WBC has a very large variance (7082932), indicating substantial spread in white blood cell counts among participants.

- **Correlation matrix:**

```
                 Age          BMI          WBC           RBC          HGB
Age       1.0000000000 -0.040876869  0.034807540 -6.865443e-03 -0.049157104
BMI      -0.0408768688  1.000000000  0.025818624  6.157955e-03  0.084277719
WBC       0.0348075399  0.025818624  1.000000000 -8.620208e-03 -0.004225359
RBC      -0.0068654430  0.006157955 -0.008620208  1.000000e+00  0.011956406
HGB      -0.0491571036  0.084277719 -0.004225359  1.195641e-02  1.000000000
Plat      0.0051278922 -0.002244307 -0.016707962  6.998979e-03 -0.035313151
AST.1    -0.0288045610  0.006063415  0.005817019 -7.787843e-03 -0.016109422
ALT.1     0.0009238417  0.040582056 -0.081999012 -1.099240e-02 -0.010845397
ALT.48    0.0316467102 -0.037562671 -0.006395362 -8.812899e-02 -0.026346571
RNA.Base  0.0107106754 -0.027644603  0.022229860 -2.087428e-02 -0.044459521
RNA.EOT  -0.0461510826 -0.001318958 -0.009649546 -3.624276e-05  0.020595547
                Plat        AST.1         ALT.1       ALT.48    RNA.Base
Age       0.005127892 -0.028804561  0.0009238417  0.031646710  0.01071068
BMI      -0.002244307  0.006063415  0.0405820562 -0.037562671 -0.02764460
WBC      -0.016707962  0.005817019 -0.0819990124 -0.006395362  0.02222986
RBC       0.006998979 -0.007787843 -0.0109923957 -0.088128995 -0.02087428
HGB      -0.035313151 -0.016109422 -0.0108453974 -0.026346571 -0.04445952
Plat      1.000000000 -0.038747514  0.0488148197 -0.022150705 -0.01951201
AST.1    -0.038747514  1.000000000  0.0569623314  0.003858614  0.00533366
ALT.1     0.048814820  0.056962331  1.0000000000  0.045826960  0.03000226
ALT.48   -0.022150705  0.003858614  0.0458269597  1.000000000  0.03498615
RNA.Base -0.019512007  0.005333660  0.0300022572  0.034986151  1.00000000
RNA.EOT   0.066319517 -0.070470302 -0.0182890163 -0.009189864  0.01220674
              RNA.EOT
Age      -4.615108e-02
BMI      -1.318958e-03
WBC      -9.649546e-03
RBC      -3.624276e-05
HGB       2.059555e-02
Plat      6.631952e-02
AST.1    -7.047030e-02
ALT.1    -1.828902e-02
ALT.48   -9.189864e-03
RNA.Base  1.220674e-02
RNA.EOT   1.000000e+00
```

Matrix_2: Correlation matrix

This matrix standardizes the relationships between variables. Each value ranges between -1 and 1, where values close to 1 indicate a strong positive correlation, values close to -1 indicate a strong negative correlation, and values near 0 suggest weak or no correlation. The diagonal elements are always 1, as they represent the correlation of each variable with itself. Most correlations are close to zero, indicating weak relationships between variables such that age and BMI have a value near to zero with correlation (-0.041), suggesting no meaningful relationship.

- **Some frequency tables for categorical variables:**

Table 2: Frequency table for Gender

Level	Freq	Perc	Cumfreq	Cumperc
1	482	49.8%	482	49.8%
2	486	50.2%	968	100.0%

Based on the data, level 1 for male and level 2 for female. So, we can conclude that we have 482 male patients in the train data represents 49.8% of the total data, and 486 female patient represents 50.2% of the total data.

Table 3: Frequency table for Fever

Level	Freq	Perc	Cumfreq	Cumperc
1	467	48.2%	467	48.2%
2	501	51.8%	968	100.0%

From data, level 1 for patient has not experienced a raised body temperature and level 2 for patient has experienced a raised body temperature. So, we can conclude that 467 of patients have experienced a raised body temperature represents 48.2% of the total data, and 502 of patients don't have experienced a raised body temperature represents 51.8% of the total data. This indicate that percentage of patient have fever is greater than half of the data.

Table 4: Frequency table for Baseline Histological Staging

Level	Freq	Perc	Cumfreq	Cumperc
1	238	24.6%	238	24.6%
2	231	23.9%	469	48.5%
3	255	26.3%	724	74.8%
4	244	25.2%	968	100.0%

From data we know that Baseline Histological Staging represents the degree of liver fibrosis at the start of treatment and level 1 for Mild Fibrosis, level 2 for Moderate Fibrosis, level 3 for Severe Fibrosis, and level 4 for Cirrhosis.

From the table, we notice that 238 patients have mild fibrosis represents 24.6**%** of the dataset, 231 patients have moderate fibrosis represents 23.9% of the dataset, 255 patients have severe fibrosis represents 26.3% of the dataset (the largest percentage among all stages), and the last 244 patients have cirrhosis represents 25.2% of the dataset

- **Contingency tables for categorical variables:**

I. Two-way Contingency table

Table 5: Two-way table for HC & Gender

HCV \ Gender	1 (Male)	2 (Female)
1	**179**	**176**
2	**303**	**310**

The two-way table for HCV status and Gender shows that out of 968 individuals, 355 are HCV-negative (179 males and 176 females) and 613 are HCV-positive (303 males and 310 females).

A Chi-Square test was used to check if there is an association between HCV status and gender. The test result 0.0535, with **p-value = 0.8171 > α = 0.05** shows no significant relationship.

II. Three-way Contingency table

Table 6: Three-way table for Hcv & Epigastric pain holding Gender

Gender	HCV	Epigastric.pain	
		1 (No pain)	2 (Pain)
Male	**1**	91	88
	2	148	155
Female	**1**	104	72
	2	138	172

The three-way table shows the distribution of individuals based on HCV status (1 = Negative, 2 = Positive), Epigastric pain (1 = No pain, 2 = Pain), and Gender (1 and 2). For Gender 1, out of 482 individuals, 303 are HCV positive, and the distribution of epigastric pain is almost even (239 without pain, 243 with pain). Similarly, for Gender 2, out of 486 individuals, 310 are HCV positive, and epigastric pain is also evenly distributed (242 without pain, 244 with pain). Overall, HCV positive cases are more common than negative in both genders, and epigastric pain appears evenly spread across all groups. This data suggests potential relationships between HCV status, epigastric pain, and gender, which could be explored further through statistical tests like Chi-Square or logistic regression.

We made the Mantel-Haenszel test, the results indicate a statistically significant association between HCV status and Epigastric pain, controlling for Gender. The test yields a **chi-squared value of 5.821** with 1 degree of freedom and a **p-value of 0.01584**, which is less than the significance level of 0.05. This suggests that the null hypothesis of no association can be rejected.
The **estimated common odds ratio is 1.39,** with a **95% confidence interval of 1.07 to 1.81.** This indicates that individuals with HCV are approximately 1.39 times more likely to report epigastric pain compared to those without HCV, adjusting for gender. Since the confidence interval does not include 1, the association is both statistically and practically significant.

- ## Plots:
  - ### Bar plots:

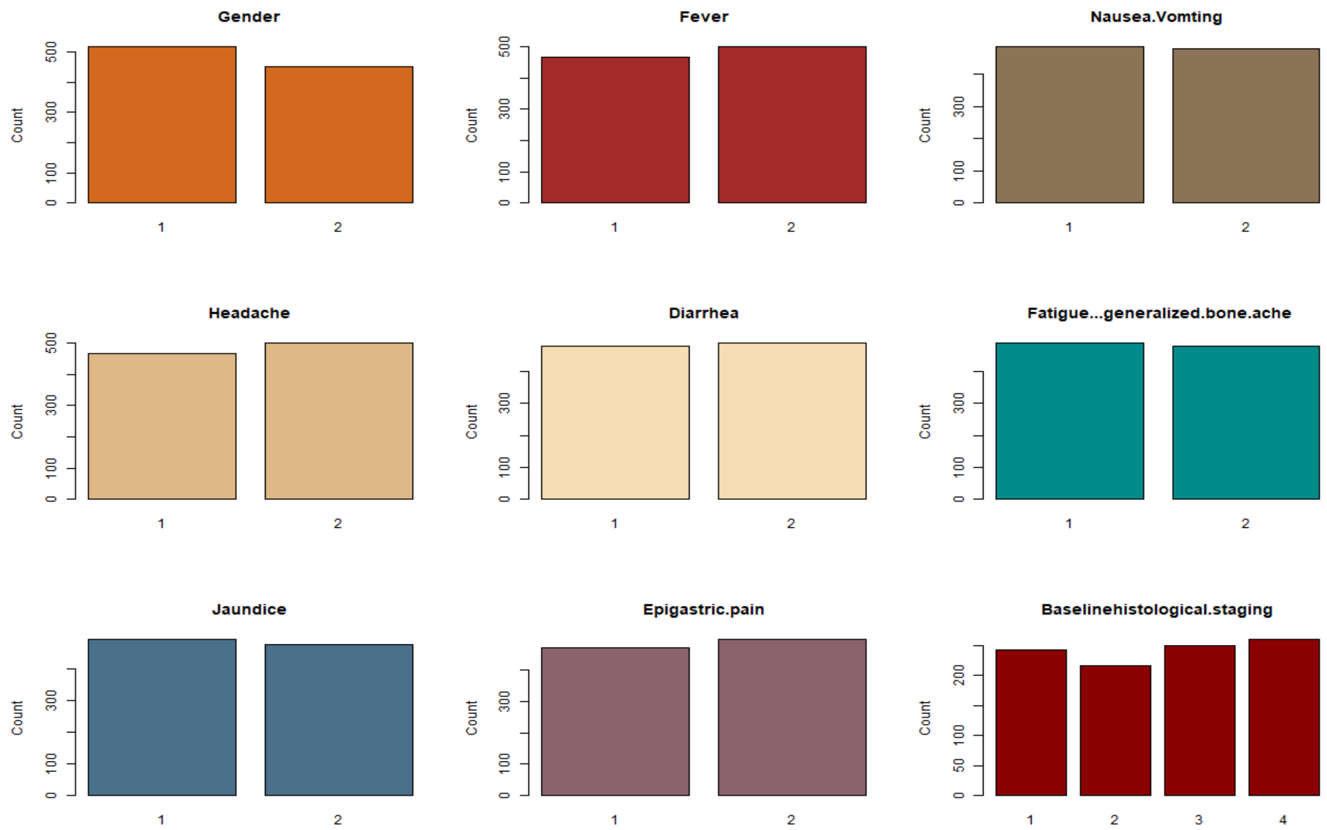

Figure 2: Bar plots for categorical variables

As it is shown from figure 2:

The first chart for gender variable shows that 1 which refers to the male gender is being the category that most of the patients belong to, and the difference between the two categories is not that much big.

The last chart for baselinehistological.staging variable which has 1, 2,3 and 4 categories, and they refer to Mild Fibrosis, Moderate Fibrosis, Severe Fibrosis, and Cirrhosis sequentialy, the chart shows that category two is the lowest one which means that the lowest number of patients their baselinehistological.staging is moderate fibrosis. The other categories are

The other plots for the other different variables show that with a fairly equal distribution of patients are involved in 1 which refers to “Absent” category and those are involved in the other category 2 which refers to “Present”. This indicates most of the other categorical variables in the chosen data are evenly distributed across the sample.

– GGpairs plot:

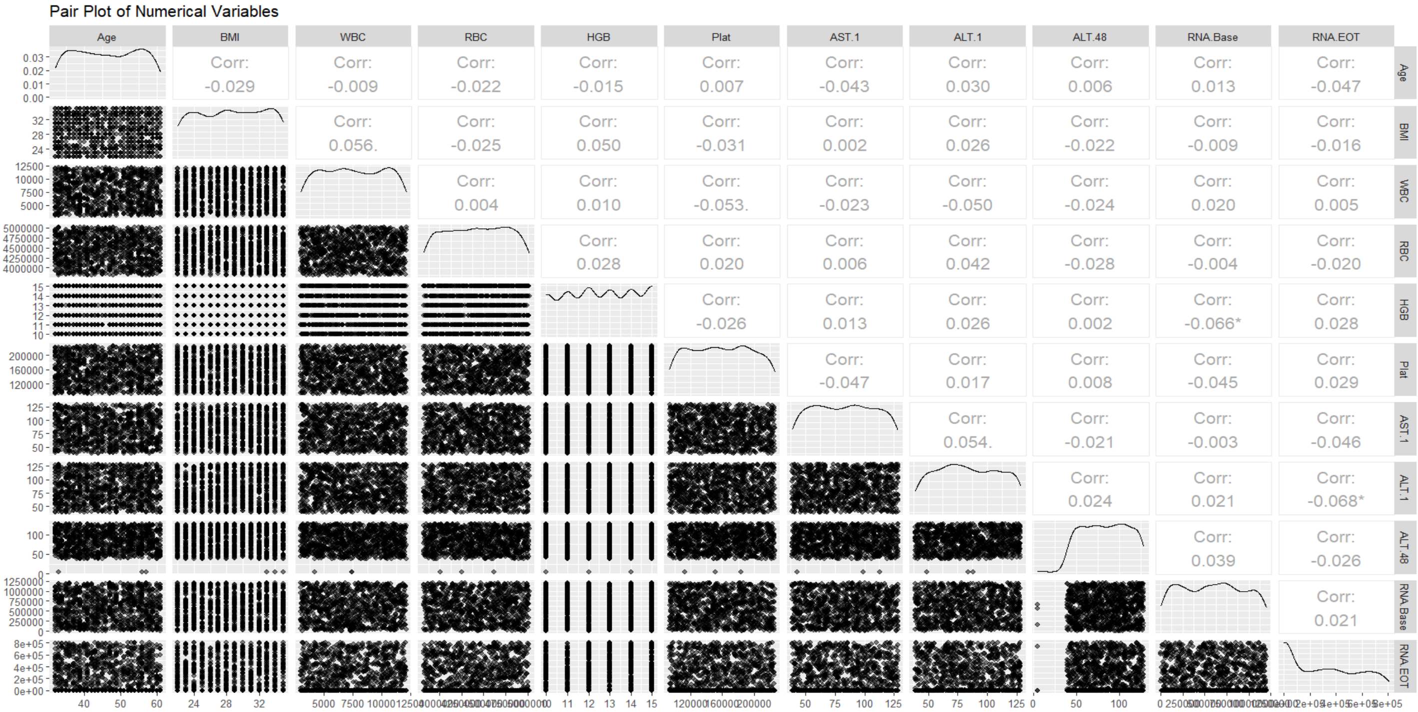

Figure 3: ggpairs for numerical variables

The plot in figure_3 provides pairwise comparisons of the variables, including scatterplots, density plots, and correlation coefficients. Most scatterplots show no strong linear relationships among the variables, as the points are widely dispersed, indicating weak or no correlation between most pairs. **We can see that the highest correlation** coefficient equals -0.068 which refers to the correlation between “ALT.1 and RNA.EOT”, and this indicates very week negative relationship between the two variables.

The other relationships are also very week, also some of them their correlation coefficient is close to zero which indicates no correlation and suggesting no meaningful relationship, like the correlation coefficient -0.003 of “AST.1 and RNA.Base”. The variables related to liver enzymes (AST.1 and ALT.1) do not show any strong correlations with other clinical measures, suggesting that these markers might act independently in this dataset.

– The heat map:

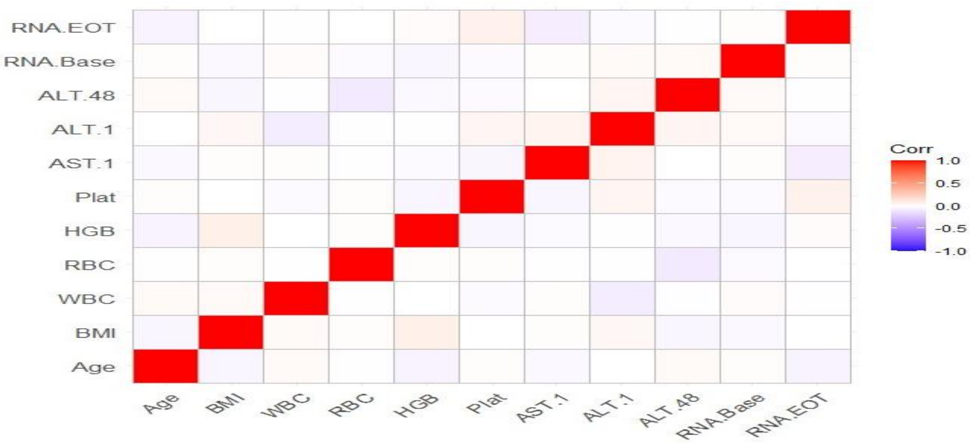

Figure 4: Correlation Heatmap

The heatmap in figure_4 presents a correlation matrix of different variables, the diagonal line of red squares indicates a perfect positive correlation between a variable and itself, as expected, since these are the variance. We can notice that the correlation between variables appear to be very low or there is no correlation.

– Density Plots for categorical vs. numerical:

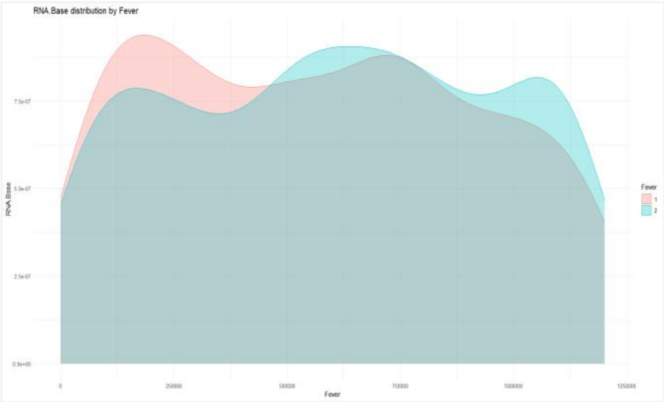

Figure 5 : RNA.Base distribution by Fever

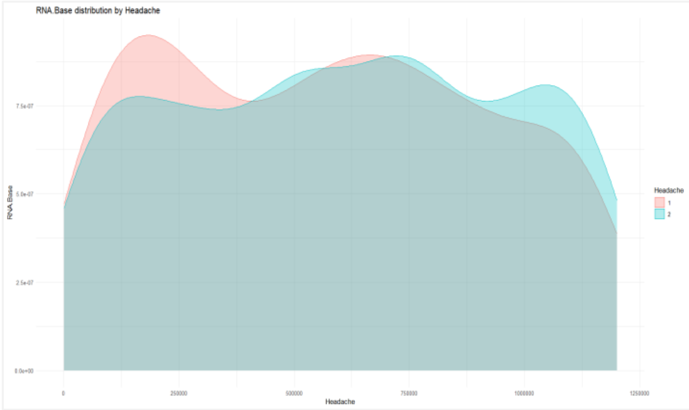

Figure 6: RNA.Base distribution by Headache

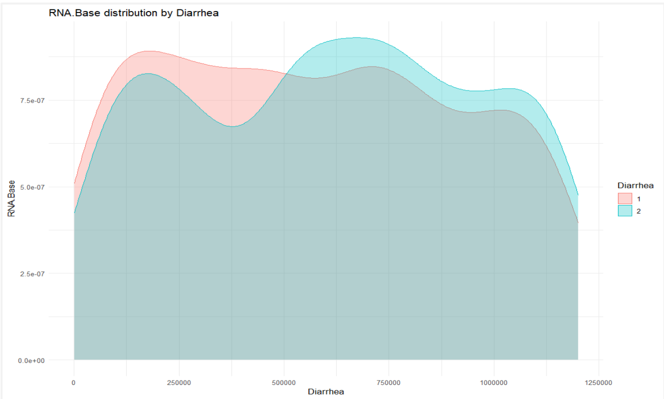

Figure 7: RNA.Base distribution by Jaundice

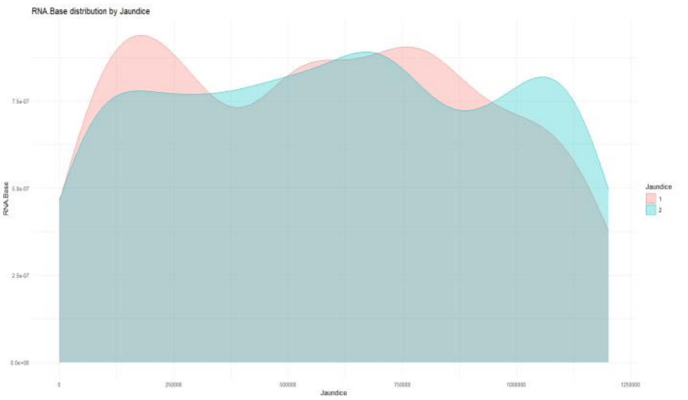

Figure 8: RNA.Base distribution by Diarrhea

The first plot in figure_5 represents the distribution of RNA.Base levels by Fever status, with two categories "1 and 2" which are "Absent and Present" sequentialy, the plot illustrates fever and its affection on RNA.Base, the red curve "Fever = 1" appears to have a peak at a lower RNA.Base level compared to the blue curve "Fever = 2".

The second plot in figure_6 represents the distribution of RNA.Base levels by Headache status, with two categories "1 and 2" which are "Absent and Present" sequentialy, the plot illustrates headache and its affection on RNA.Base, the red curve "Headache = 1" appears to have a peak at a lower RNA.Base level compared to the blue curve "Headache = 2".

The third plot in figure_7 represents the distribution of RNA.Base levels by Jaundice status, with two categories "1 and 2" which are "Absent and Present" sequentialy, the plot illustrates Jaundice and its affection on RNA.Base, The red curve "Jaundice = 1" and the blue curve "Jaundice = 2" show that there is some overlap between the distributions of the two groups, which suggests that the RNA.Base levels in both Jaundice categories share similar ranges, and the differences in the shapes and peaks of the two distributions might suggest a biological or clinical distinction in RNA Base levels between the jaundice categories.

The last plot in figure_8 represents the distribution of RNA.Base levels by Diarrhea status, with two categories "1 and 2" which are "Absent and Present" sequentialy, the plot illustrates Diarrhea and its affection on RNA.Base, The red curve "Diarrhea = 1" has its peak slightly earlier than the blue curve "Diarrhea = 2", suggesting a difference in the distribution of RNA.Bases between the two groups, There is an overlapping region where both curves share a similar distribution, but their peaks are distinct.

– Pie charts for categorical variables:

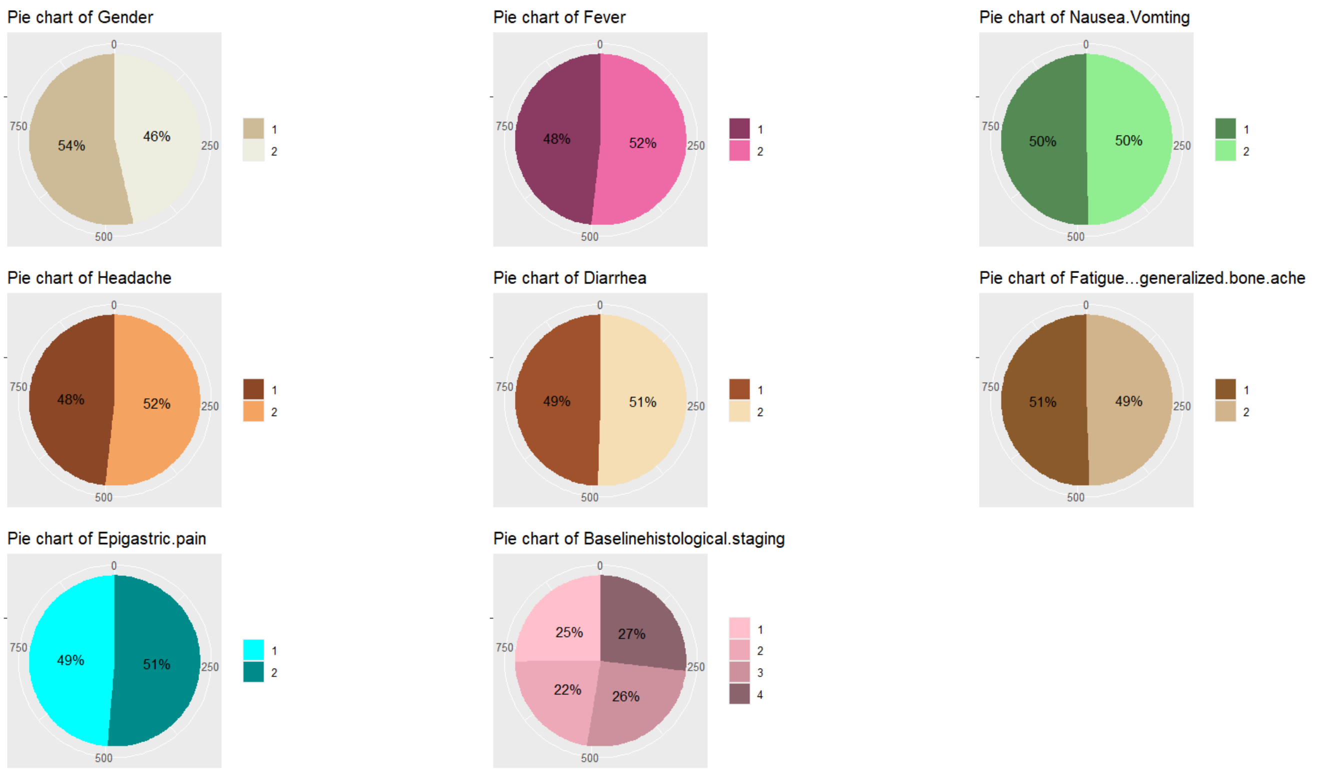

Figure 9: Pie chart for categorical variables

From figure_9 we notice that:

The first chart shows gender variable, where 54% of the patients belong to male category, compared to 46% female category, where male: 1, and female: 2.

The second chart shows fever variable which indicates whether the patient has experienced a raised body temperature, where 48% of the patients belong to Absent category, compared to 52% Present category, where Absent: 1, and Present: 2.

The third chart shows Nausea.Vomiting variable which represents whether the patient has experienced nausea and/or vomiting, where 50% of the patients belong to Absent category, compared to 50% Present category, where Absent: 1, and Present: 2.

The fourth chart shows Headache variable which indicates whether the patient has experienced headaches, where 48% of the patients belong to Absent category, compared to 52% Present category, where Absent: 1, and Present: 2.

The fifth chart shows Diarrhea variable which indicates whether the patient has experienced diarrhea, where 49% of the patients belong to Absent category, compared to 51% Present category, where Absent: 1, and Present: 2.

The sixth chart shows Fatigue & Generalized Bone Ache variable which indicates whether the patient has experienced fatigue and/or bone aches, where 51% of the patients belong to Absent category, compared to 49% Present category, where Absent: 1, and Present: 2.

The seventh chart shows Epigastric.pain variable which refers to pain or discomfort in the upper central region of the abdomen, directly above the stomach, where 49% of the patients belong to Absent category, compared to 51% Present category, where Absent: 1, and Present: 2.

The last chart shows Baseline Histological Staging variable which represents the degree of liver fibrosis at the start of treatment, assessed through liver biopsy. Fibrosis is the accumulation of scar tissue in the liver due to chronic inflammation, this variable is often scored on a scale from **1** to **4**, where 25% of the patients belong to Mild Fibrosis category, 22% belong to Moderate Fibrosis category, 26% belong to Severe Fibrosis category, and 27% belong to Cirrhosis category, where Mild Fibrosis: 1, Moderate Fibrosis: 2, Severe Fibrosis: 3, and Cirrhosis: 4.

Overall we can notice that the patients do not belong to a specific category, they are distributed evenly.

- **Scatter Plot of RNA.Base vs RNA.EOT:**

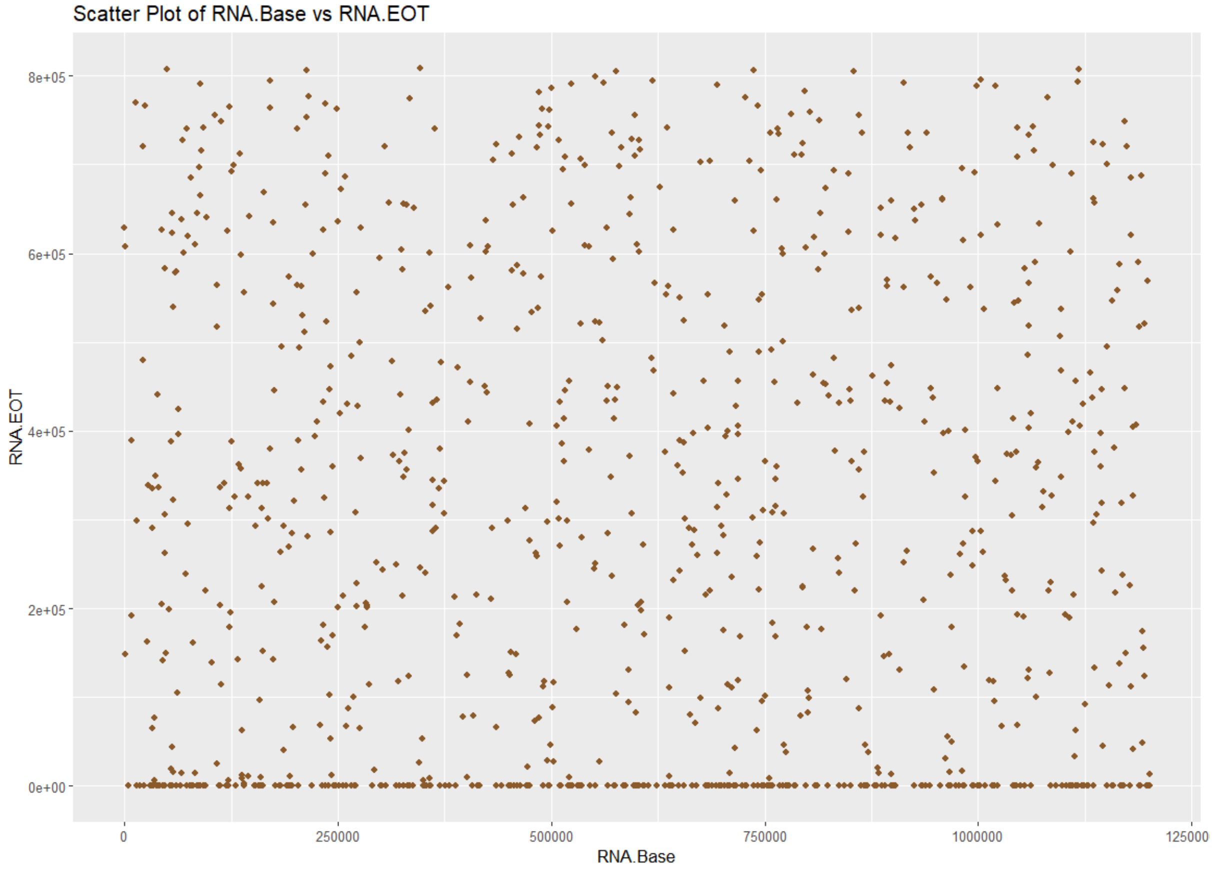

Figure 10: Scatter Plot of RNA.Base vs RNA.EOT

The scatter plot in figure_10 suggests that the two variables are well-separated, and random

The scatter plot in figure_10 illustrates the relationship between RNA.Base levels and RNA.EOT levels, while the x-axis represents the RNA.Base values, and the y-axis corresponds to RNA.EOT values. From the plot, we can notice that a large proportion of the data points are clustered at the lower end of the y-axis, indicating that many individuals have low RNA.EOT levels. However, RNA.Base values appear more uniformly distributed across the x-axis. This pattern might suggest that RNA.Base levels are not strongly correlate with RNA.EOT levels, or there might be other factors influencing the outcome.

– Histogram for numeric variable:

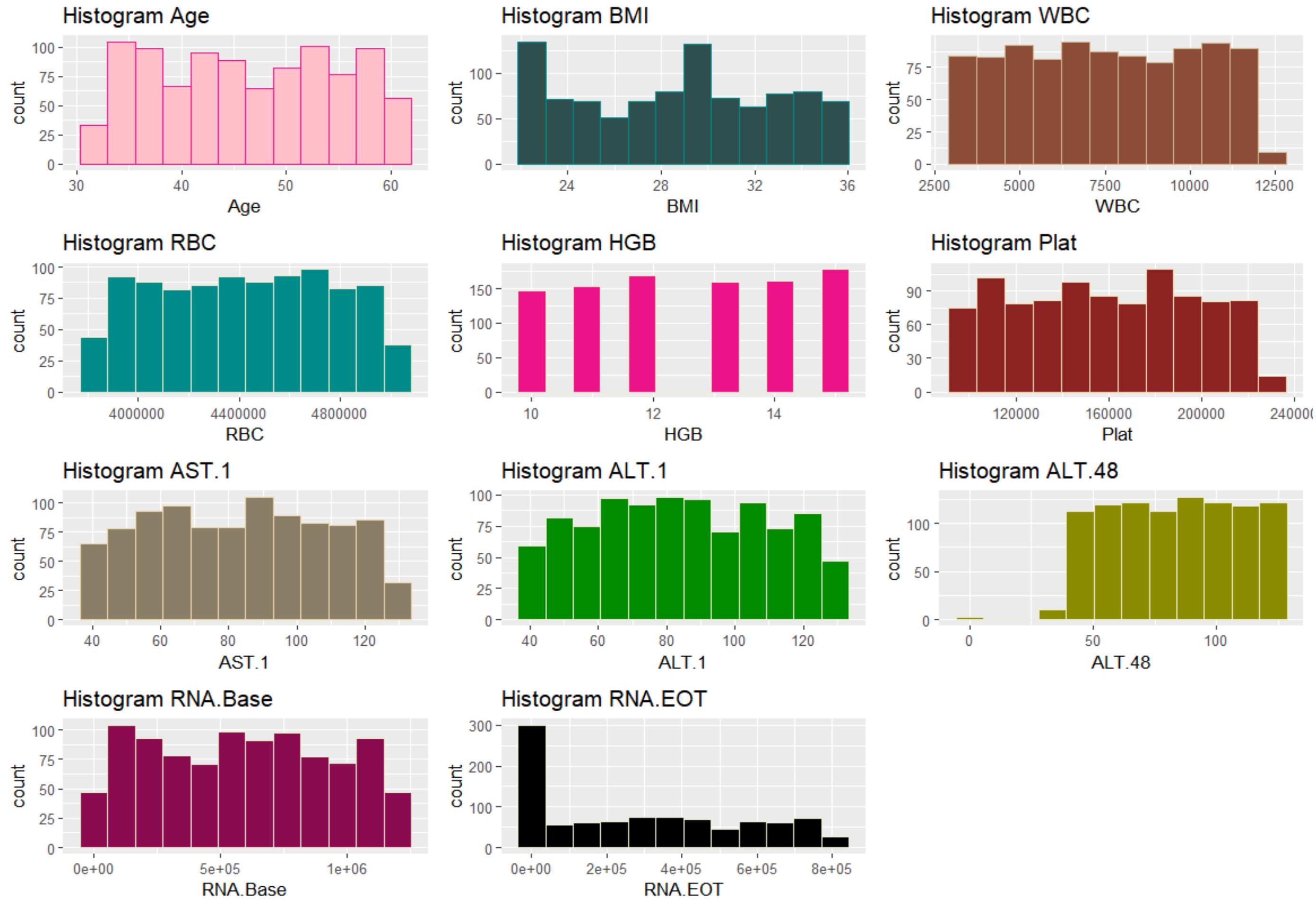

Figure 11: Histogram for numeric variable

From figure_11 we can notice that:

The Age histogram shows a relatively uniform distribution across the age ranges, with a slight peak in the 50-60 age group.

For WBC (White Blood Cell count), the histogram shows a fairly even distribution across the range, though there is a slight decrease in frequency for the higher WBC values.

The RBC (Red Blood Cell count) histogram has a relatively uniform distribution, showing no significant peaks or valleys. This implies a balanced representation of RBC values across the population.

The HGB (Hemoglobin) histogram has a symmetric distribution centered around 12-14, suggesting that most individuals have hemoglobin levels within a normal range.

The Plat (Platelet count) distribution shows moderate variation across the range, with a few peaks and dips.

The AST.1 histogram shows an approximately uniform distribution, though the counts drop slightly at the higher end of the range.

For ALT.48, the distribution is highly skewed, with most values concentrated near zero and a steep drop-off for higher levels.

Finally, **the RNA.EOT** distribution is heavily skewed, with most observations near zero. There are very few individuals with higher RNA levels at the end of treatment.

- ## Analysis:

To group Hepatitis C patients based on their demographic, clinical, and comorbidity profiles, we applied hierarchical and K-means clustering. Initially, variables with high variation were log-transformed to reduce skewness and ensure a more balanced scale. The dataset was then standardized, and a distance matrix was created using Euclidean distance. Hierarchical clustering with Ward's method was performed, and a dendrogram was plotted to explore the data structure. Both the dendrogram and the elbow plot indicated two optimal clusters, which were then confirmed using K-means clustering for more efficient segmentation. Cluster 1 included patients with lower RNA-related variables (indicating better treatment responses, they almost do not have virus c after treatment), while Cluster 2 showed higher RNA values, suggesting more severe disease progression (indicating that patients still have the virus). We correctly saved this clusters in our data and we will use these clusters as the outcome variable (y) in the analyses for our project.

- ## Logistic regression model:

Binary Logistic regression is a statistical method used for binary classification "where the outcome variable is categorical with two classes, typically coded as 0 and 1". logistic regression predicts the probability of the outcome belonging to one of the two classes. The model estimates the relationship between the dependent binary variable and one or more independent variables using the logistic function.

Logistic regression uses one or more predictor variables to estimate the probability of an event occurring. It first calculates the log-odds, which is the logarithm of the ratio of the probability of one category to the other. These log-odds are then transformed into a probability value between 0 and 1 using the sigmoid function. If the calculated probability is greater than 50% (0.5), the model predicts the event as occurring (1), and if it's below 50%, the event is predicted as not occurring (0). The model's goal is to find the best combination of predictors that explains the likelihood of the event.

To predict whether the patient still has c virus after the end of treatment or not, we will use binary logistic regression since our response variable is categorical.

First, we will apply initial logistic model and then we will do stepwise to identify the most important predictors for HCV status.

Table 7: Stepwise

Coefficients	Estimate	Std. Error	z value	Pr (>|z|)	Exponential Coeff	VIF
Intercept	**-2.984e-01**	**8.133e-01**	**-0.367**	**0.7137**	**0.7420098**	
Age	**-2.691e-02**	**1.200e-02**	**-2.243**	**0.0249**	**0.973**	**1.027934**
Fever2	**4.566e-01**	**2.111e-01**	**2.163**	**0.0306**	**1.5787016**	**1.020596**
Fatigue	**-3.557e-01**	**2.108e-01**	**-1.687**	**0.0915**	**0.7006686**	**1.024188**
Epigastric.pain2	**5.012e-01**	**2.110e-01**	**2.375**	**0.0176**	**1.6506286**	**1.029343**
AST.1	**-6.073e-03**	**4.126e-03**	**-1.472**	**0.1410**	**0.9939454**	**1.028953**
ALT.1	**-7.275e-03**	**4.154e-03**	**-1.751**	**0.0799**	**0.9927517**	**1.035622**
ALT.48	**7.292e-03**	**4.046e-03**	**1.802**	**0.0715**	**1.0073188**	**1.022609**
RNA.EOT	**1.283e-05**	**8.657e-07**	**14.825**	**<2e-16**	**1.0000128**	**1.101751**

The stepwise logistic regression model helped simplify the analysis with number of Fisher Scoring iterations = 7, by selecting the most important variables for predicting HCV status. The final model included Age, Fever, Fatigue, Epigastric Pain, AST.1, ALT.1, ALT.48, and RNA.EOT. Among these, Age, Fever, Epigastric Pain, and RNA.EOT were statistically significant, with RNA.EOT being the strongest predictor **(p < 2e-16).** Some variables like Fatigue, AST.1, ALT.1, and ALT.48 were less significant but still included in the model. The model's residual deviance decreased significantly **(from 1272.33 to 589.09),** and the AIC was **607.09**, showing that the model fits the data well. Overall, the stepwise method helped create a simpler and more effective model by focusing on the most important predictors.

The exponentiated coefficients from the stepwise logistic regression model show how each variable affects the odds of HCV. Age has an odds ratio of 0.973, meaning the odds of HCV decrease by about 2.7% with every extra year. Fever increases the odds by 1.58 times, while fatigue or generalized bone ache reduces the odds by 30%. Epigastric pain raises the odds by 1.65 times. Lab measures like AST.1 and ALT.1 have minimal effects, slightly lowering the odds, while ALT.48 slightly increases it. RNA.EOT, although significant, has a very small effect. These results show how different factors contribute to predicting HCV.

The VIF values for the stepwise model are all close to 1, which means there is little to no multicollinearity between the variables. This shows that the predictors in the model don't interfere with each other and are reliable.

- **Binary logistic regression model:**

$$\log \frac{\widehat{\text{Target}}}{1-\widehat{\text{Target}}} = \widehat{\beta}_0 + \widehat{\beta}_1 \text{Age} + \widehat{\beta}_2 \text{ Fever2} + \widehat{\beta}_3 \text{ Epigastric.pain2} + \widehat{\beta}_4 \text{ RNA.EOT} + \widehat{\beta}_5 \text{AST.1} + \widehat{\beta}_6 \text{ALT.1} + \widehat{\beta}_7 \text{ ALT.48} + \widehat{\beta}_8 \text{ Fatigue...generalized.bone.ache.}$$

Table 8: ANOVA test

Coefficients	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
NULL	1		967	1272.33	
Age	1	6.70	966	1265.63	0.009639
Fever	1	1.85	965	1263.78	0.173694
Fatigue	1	0.22	964	1263.56	0.638266
Epigastric.pain	1	7.00	963	1256.55	0.008134
AST.1	1	3.41	962	1253.15	0.064942
ALT.1	1	0.68	961	1252.46	0.408224
ALT.48	1	1.03	960	1251.43	0.309356
RNA.EOT	1	662.34	959	589.09	< 2.2e-16

The ANOVA table for stepwise logistic regression model shows the contribution of each variable to reducing the model's deviance. Significant variables with smaller p-values (Age, Epigastric Pain, and RNA.EOT) have the strongest association with the outcome (HCV). Specifically, RNA.EOT stands out as the most influential variable **(p-value = 2.2e-16),** causing a substantial drop in residual deviance. Variables like Fever, Fatigue, ALT.1, and ALT.48 show higher p-values, suggesting weaker contributions. This analysis confirms the importance of certain predictors while identifying less impactful ones.

For predicting the log-odds of the test data using the stepwise logistic regression model and converts these into probabilities using the logistic function. The summary of the predicted probabilities shows that the minimum predicted probability is 0.04231, indicating a very low chance of having HCV in some cases. About 25% of the predictions are below 0.20, while the median probability is 0.80755, meaning half of the predictions are above this value, suggesting a higher likelihood of having HCV. The average probability is 0.63102, which is slightly above 0.5, showing the model tends to predict a positive outcome more often. Around 75% of the predictions are above 0.99, indicating strong confidence in predicting the positive class for some cases, with the maximum probability being 0.99993. This indicates that the model has some cases with a high likelihood of being positive for HCV and others with low likelihood, so a threshold = 0.5 could be used to classify the predictions into binary outcomes.

Table 9: Confusion Matrix

Prediction	Reference		
	0	1	Total
0	130	26	156
1	31	229	260
Total	161	255	416

The confusion matrix for the test data shows how well the model predicted HCV status. Out of 130 true negatives (0), the model correctly predicted 130 as 0 but made 26 wrong predictions as 1. Out of 260 true positives (1), the model correctly predicted 229 as 1 but misclassified 31 as 0. Overall, the model did a good job, but there were still some mistakes, especially in predicting negatives (0). We can calculate accuracy, sensitivity, and specificity to see how well the model performed.

Table 10: Performance & Goodness of fit

Accuracy	Sensitivity	Specificity	Balanced Accuracy	McFadden
0.8629808	**0.8807692**	**0.8333333**	**0.8527**	**0.5444566**

The model's performance is quite strong, with an accuracy of 86.3%, meaning it correctly predicted HCV status most of the time. The 95% confidence interval for accuracy ranges from 82.6% to 89.4%. The model also has good sensitivity (80.8%), which shows its ability to correctly identify positive cases, and specificity (89.8%), which shows its ability to correctly identify negative cases. The Kappa value of 0.7096 indicates substantial agreement between the predicted and actual values. The positive predictive value (PPV) is 83.3%, and the negative predictive value (NPV) is 88.1%, suggesting the model is reliable in predicting both positive and negative cases. The balanced accuracy is 85.3%, which accounts for the balance between sensitivity and specificity. The P-value for the McNemar's test is 0.5962, which suggests there's no significant difference between the number of false positives and false negatives.

The McFadden pseudo-R^2 for the logistic regression model was calculated to be 0.537. This value indicates that the model explains approximately 53.7% of the variance in the outcome variable, which is considered a good fit. McFadden's R^2 is a measure of model goodness-of-fit, with values closer to 1 representing better explanatory power. In this case, a value of 0.537 suggests that the model provides a substantial amount of insight into the relationship between the predictor variables and the outcome, though there may still be other factors influencing the result.

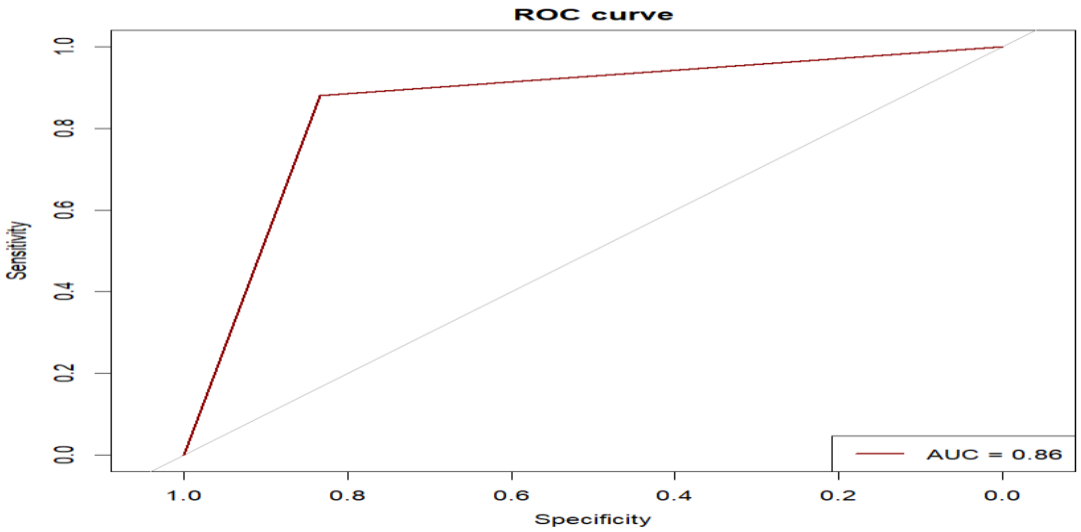

Figure 12: Roc Curve

The ROC curve for the model shows an AUC of 0.86, which means the model does a good job of distinguishing between positive and negative cases. It has high sensitivity, correctly identifying most of the positive cases, and good specificity, accurately classifying many of the negative cases. Overall, the ROC curve indicates that the model performs well in classifying both types of cases.

▪ Machine learning algorithms:

Machine learning algorithms are pieces of code that help people explore, analyze, and find meaning in complex data sets. Each algorithm is a finite set of unambiguous step-by-step instructions that a machine can follow to achieve a certain goal.

We will study two types of Machine learning: Decision tree and K-Nearest Neighbors.

I. Decision tree:

A decision tree splits data into branches based on feature values, creating a tree-like structure. So we'll apply it to classify whether a patient has hepatitis C Virus or not.

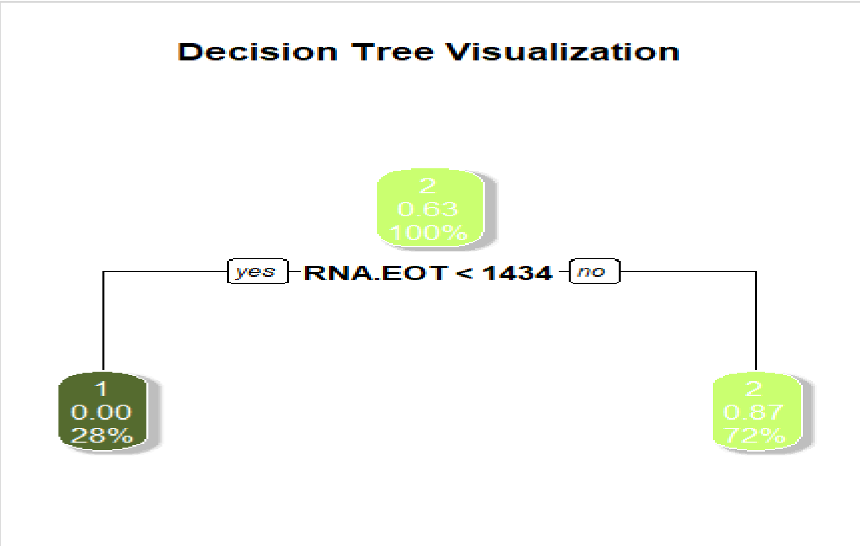

Figure 13: Decision tree for the model

The decision tree model was developed using the rpart package in R to classify observations based on the target variable (HCV).

In this initial stage, 355 observations are misclassified, which is equivalent to 36.6%, and 613 are correctly classified, which is equivalent to 63.4%, meaning that out of 100 individuals, about 36 individuals are misclassified and about 64 are correctly classified.

Table 11: Error table resulting from decision tree

		1	2
RNA.EOT< 1434	1	267 100%	0 0%
RNA.EOT>=1434	2	701 12.5%	88 87.5%

The first split in the tree occurs based on the variable RNA.EOT (RNA Level at End of Treatment) with a threshold value of 1434. Observations with RNA.EOT values less than 1434 are clustered in the left sub node. This node contains 267 observations, all of which are classified as Class 1, i.e., those who were cured of the virus after taking the treatment with 100% probability, indicating perfect classification for this group.

On the other hand, observations where RNA.EOT is greater than or equal to 1434 are grouped into the right child node. This node contains 701 observations, with 88 misclassified cases. The majority class in this group is Class 2, i.e., those who have not been cured of the virus after taking treatment, with probabilities of 12.55% for Class 1 and 87.45% for Class 2. This indicates that most observations in this node are correctly classified as Class 2.

– The most important variable in splitting the data

Table 12: Importance of the variables in splitting the data

Variable	Overall
RNA.EOT	295.711920
Age	5.732580
WBC	3.919924
AST.1	3.205623
ALT.1	3.137685
Gender	0.000000
BMI	0.000000
Fever	0.000000
Nausea.Vomting	0.000000
Headache	0.000000
Diarrhea	0.000000
Fatigue...generalized.bone.ache	0.000000
Jaundice	0.000000
Epigastric.pain	0.000000
RBC	0.000000
HGB	0.000000
Plat	0.000000
ALT.48	0.000000
RNA.Base	0.000000
Baselinehistological.staging	0.000000

From Table 12, we can conclude that: the variable importance analysis reveals that RNA.EOT is the most influential variable in the decision tree model, with a significantly high importance score of 295.71. This indicates that RNA.EOT plays a critical role in splitting the data and determining the classification outcome.

Other variables, such as Age, WBC (white blood cell count), AST.1, and ALT.1, also contribute to the model but to a much lesser extent, with importance scores of 5.73, 3.92, 3.21, and 3.14, respectively. Despite its impact, it is very small, almost non-existent, and this is clear in the decision tree model.

several variables, including Gender, BMI, Fever, Nausea.Vomiting, Diarrhea…. etc., have an importance score of 0. This suggests that these features were not used in the decision-making process of the tree and did not contribute to splitting the data.

– Confusion matrix

Table 13: confusion matrix.

		Reference		
		1	2	Total
Prediction	1	117	39	156
	2	1	259	260
	Total	118	298	416

Table 13 shows the prediction in the test data, and it is clear that it predicted the presence of 117 people who were not cured after taking the treatment, and indeed these people were cured after receiving the treatment.

It was predicted that there was one person who was cured after taking the treatment, but in reality he was not cured.
There were 39 individuals who were predicted not to be cured after taking the treatment, but in fact they were cured.
It was also predicted that 259 people were cured after taking the treatment, and as predicted, they were cured.

Table 11: accuracy measures

Measure	Value
Accuracy	0.9038
95% CI	(0.8714, 0.9304)
No Information Rate	0.7163
P-Value [Acc > NIR]	< 2.2e-16
Kappa	0.7844
Mcnemar's Test P-Value	4.909e-09
Sensitivity	0.9915
Specificity	0.8691
Pos Pred Value	0.7500
Neg Pred Value	0.9962
Prevalence	0.2837
Detection Rate	0.2812
Detection Prevalence	0.3750
Balanced Accuracy	0.9303

The confusion matrix and related statistics provide an evaluation of the model's performance on the test dataset. The overall accuracy of the model is 90.38%, with a 95% confidence interval ranging from 87.14% to 93.04%. This indicates that the model classifies most observations correctly.

The Kappa statistic is 0.7844, suggesting a strong agreement between the predicted and actual classifications beyond what would be expected by chance.

The sensitivity (true positive rate) is 99.15%, meaning the model correctly identifies the majority of positive cases. The specificity (true negative rate) is 86.91%, indicating a slightly lower ability to identify negative cases.

The positive predictive value (precision) is 75%, while the negative predictive value is exceptionally high at 99.62%, showing the model's strong reliability in predicting negative cases.

Overall, these results highlight the model's effectiveness, particularly in identifying positive cases, while maintaining reasonable performance in distinguishing negative cases.

II. K-Nearest Neighbors (KNN)

KNN is a simple algorithm that predicts the output for a new data point based on the similarity (distance) to its nearest neighbors in the training dataset, used for both classification and regression tasks.

To prepare the data, numerical features were extracted from both the training and testing datasets. These features were then standardized using the scale function to ensure that all variables contribute equally to the distance calculations and that there was no difference in the units of measurement of the variables.

– Choosing K:

To identify the optimal value of k for the k-Nearest Neighbors (KNN) classifier, a range of values from 1 to 15 was tested, and the misclassification error was computed for each value.

To optimize the number of neighbors (k) for the k-Nearest Neighbors (KNN) classifier, a range of values from 1 to 15 was tested, and the misclassification error was calculated for each value.

Table 12: Misclassification for different K

Number of K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
MisClassError	25.9	28.1	22.3	23	21.3	21.8	21.1	23	23.7	22.5	22.3	20.4	21.8	21.8	21.6

The analysis revealed that the lowest misclassification error occurred when k = 12, with an error rate of 20.43%. This makes k = 12 the optimal value for this dataset.

After determining the optimal k, the model was retrained using k = 12 and evaluated on the test dataset. The misclassification error for this configuration was approximately 21.87%, corresponding to an overall accuracy of 78.13%. This indicates that the classifier performs reasonably well at distinguishing between classes.

– Confusion Matrix:

$$\text{HCV} \begin{cases} 0 & \text{Treated after the end of treatment} \\ 1 & \text{Not treated after the end of treatment} \end{cases}$$

Table 16 Confusion Matrix

		Reference		
		0	**1**	**Total**
Prediction	**0**	96	60	156
	1	31	229	260
	Total	127	289	416

The confusion matrix provides insight into the performance of the KNN model with k = 12 on the test dataset. The matrix reveals that the model correctly classified 96 instances as Class 0 and 229 instances as Class 1, indicating a strong ability to identify true positives and true negatives. However, the model also misclassified 60 instances of Class 0 as Class 1 and 31 instances of Class 1 as Class 0.

These results highlight that the model has a higher rate of false positives (60) compared to false negatives (31). This suggests that while the model is effective at identifying positive cases, it occasionally misclassifies negative cases as positive.

▪ Functions:

The first function, **My_Function1**, calculates the difference between two key RNA-related variables, RNA.Base and RNA.EOT, which represent the baseline RNA levels and RNA levels at the end of treatment, respectively. This difference provides insight into how much the viral load has decreased during treatment, helping to evaluate the effectiveness of the therapy.

The second function, **My_Function2**, calculates the average value of a specified variable (RNA.EOT) for two categories of Hepatitis C Virus (HCV) patients: those who had a good response to treatment (category 1: no virus) and those who still have the virus (category 2). This function helps compare treatment outcomes between these groups, offering a clearer understanding of how different patient categories respond to therapy. Together, these functions are designed to analyze treatment efficacy and patient outcomes based on RNA levels.

▪ Recommendation:

Treatments should focus on reducing RNA levels, especially RNA.EOT, as it is the most important factor in disease progression and treatment response. Direct-acting antivirals (DAAs) are the best option.

▪ Conclusion:

The project explores the analysis of a dataset on Hepatitis C Virus (HCV) treatment in Egyptian patients. It uses demographic, clinical, and laboratory variables to understand factors influencing HCV outcomes and evaluates treatment effectiveness. The dataset includes 1,384 observations and 29 variables, narrowed to 20 key features (11 numerical and 9 categorical). Key numerical variables include Age, BMI, and RNA levels, while categorical variables like Gender and symptoms (Fever, Fatigue) provide additional context. The data preparation process confirmed no missing values, duplicates, or outliers.

Descriptive statistics revealed that most variables are symmetric, with varying degrees of dispersion, particularly RNA levels showing high variability. Correlation analysis indicated weak relationships among most variables. Frequency tables showed that the dataset is balanced by gender (49.8% male, 50.2% female) and presents a range of liver fibrosis stages, with severe fibrosis being the most frequent. Contingency table analyses found no significant association between HCV status and gender ($p = 0.8171$), while the Mantel-Haenszel test indicated a significant association between HCV status and Epigastric pain when controlling for gender ($p = 0.01584$).

In the logistic regression model, we first preprocessed the data by ensuring there were no missing values or outliers, and converting categorical variables (such as Gender and Fever) into factors. The target variable, HCV, was then transformed into a binary classification (0 & 1) to represent the two classes. We built the logistic regression model using R’s glm () function with binary logistic regression, including various predictor variables like Age, Gender, BMI, and others, and ensured the model converged within 100 iterations. The model was evaluated by reviewing the coefficients and significance of predictors, performing a Chi-square test, and predicting log-odds for the test dataset. We then transformed the log-odds to probabilities and classified them into binary outcomes. To assess performance, we used a confusion matrix, calculated accuracy, sensitivity, specificity, McFadden’s R^2, and checked for multicollinearity using the Variance Inflation Factor. Additionally, we plotted the ROC curve and calculated the AUC to evaluate the model's discriminative ability.

Machine learning techniques were applied, including decision trees and K-Nearest Neighbors (KNN). The decision tree identified RNA levels at the end of treatment (RNA.EOT) as the most influential variable, achieving 90.38% accuracy with strong sensitivity (99.15%) and balanced specificity (86.91%). The KNN algorithm identified the optimal number of neighbors ($k = 12$), yielding 78.13% accuracy, with a higher false positive rate compared to the decision tree.

- **Appendix:**

```r
#Packages & libraries
install.packages("caTools")
library(dplyr)
library(psych) #for descriptive statistics
library(ggcorrplot) #for correlation plot matrix
library(randomForest)  #To implement the Random Forest model
library(rpart)  #for model Decision tree
library(rpart.plot)  #for visualize the decision tree
library(caret)  #finding importante variables in spliting the data
library(car) #To display VIF to check multicollinearity
library(pROC) #To plot ROC curve
library(pscl) #To display McFadden's R²
library(class)    # For KNN algorithm
library(caTools)  # For data splitting
library(GGally)
library(ggplot2)
library(tidyr)
library(DescTools)
library(kableExtra)
library(RColorBrewer)
library(reshape2)
library(vcd)
library(Rmisc)
library(plotly)
library(gridExtra)
library(ggridges)
library(descr)
library(ggpie)
library(plyr)
library(scales)

#read data
data <- read.csv("C:/yasmeen/Data Science/Project/HCV-EGY-DATA.csv")
data <- read.csv("HCV-EGY-DATA.csv")
str(data)

#Selecting our own sample, train data (70%) which will use it and test (30%)
set.seed(123)
set.seed(42)
nrow(data)
sample_index <- sample(1:nrow(data), size = 0.7 * nrow(data))
train <- data[sample_index, ]
test <- data[-sample_index, ]
# Save the training and testing data to CSV files
write.csv(train, "train_data.csv", row.names = FALSE)
train <- read.csv("train_data.csv")
write.csv(test, "test_data.csv", row.names = FALSE)
test <- read.csv("test_data.csv")

#Check the number of observations and variables
num_observations <- nrow(train)
num_observations
num_variables <- ncol(train)
num_variables
```

```r
#Create subset of the data set with only the selected variables
train_data <- subset(train, select = c(Age, Gender, BMI, Fever, Nausea.Vomting, Headache, Diarrhea,
                                  Fatigue...generalized.bone.ache, Jaundice, Epigastric.pain, WBC,
                                  RBC, HGB, Plat, AST.1, ALT.1, ALT.48, RNA.Base, RNA.EOT,
                                  Baselinehistological.staging, HCV))
test_data <- subset(test, select = c(Age, Gender, BMI, Fever, Nausea.Vomting, Headache, Diarrhea,
                                  Fatigue...generalized.bone.ache, Jaundice, Epigastric.pain, WBC,
                                  RBC, HGB, Plat, AST.1, ALT.1, ALT.48, RNA.Base, RNA.EOT,
                                  Baselinehistological.staging, HCV))
class(train_data)
str(train_data)
View(train_data)
View(test_data)

#Cleaning & preparing data
#Convert variable types to the correct types
train_data$Gender<- as.factor(train_data$Gender)
train_data$Fever<- as.factor(train_data$Fever)
train_data$Nausea.Vomting<- as.factor(train_data$Nausea.Vomting)
train_data$Headache<- as.factor(train_data$Headache)
train_data$Diarrhea<- as.factor(train_data$Diarrhea)
train_data$Fatigue...generalized.bone.ache<- as.factor(train_data$Fatigue...generalized.bone.ache)
train_data$Jaundice<- as.factor(train_data$Jaundice)
train_data$Epigastric.pain<- as.factor(train_data$Epigastric.pain)
train_data$Baselinehistological.staging<- as.factor(train_data$Baselinehistological.staging)
train_data$HCV<- as.factor(train_data$HCV)
str(train_data)

test_data$Gender<- as.factor(test_data$Gender)
test_data$Fever<- as.factor(test_data$Fever)
test_data$Nausea.Vomting<- as.factor(test_data$Nausea.Vomting)
test_data$Headache<- as.factor(test_data$Headache)
test_data$Diarrhea<- as.factor(test_data$Diarrhea)
test_data$Fatigue...generalized.bone.ache<- as.factor(test_data$Fatigue...generalized.bone.ache)
test_data$Jaundice<- as.factor(test_data$Jaundice)
test_data$Epigastric.pain<- as.factor(test_data$Epigastric.pain)
test_data$Baselinehistological.staging<- as.factor(test_data$Baselinehistological.staging)
test_data$HCV<- as.factor(test_data$HCV)

# Check for missing values
#check if we have any missing in the data
any(is.na(train_data)) #we don't have missing values

#check outliers by boxplot
#First i will make a data frame that contains only numeric variables
#and another data frame contains only categorical variables
numeric_variables<- train_data[sapply(train_data, is.integer)]
summary(numeric_variables)
categorical_variables<- train_data[sapply(train_data, is.factor)]

#we will create a for loop to create boxplots for each variable
par(mfrow = c(3,4))  # 3 rows and 4 columns
colors <- c("steelblue4", "violetred1", "seagreen", "turquoise1", "mediumorchid4", "chocolate1", "darkred",
            "gold", "tan4", "purple", "tomato")
i<- 1
for (var_name in names(numeric_variables)) {
  boxplot(numeric_variables[[var_name]],
          main = paste("Boxplot of", var_name),
          col = colors[i])
  i<- i+1} #we notice that there is no outliers
layout(1)

#check for duplicates & Remove if we have any.
duplicate<- sum(duplicated(train_data))
duplicate #there is no duplicate
```

```r
#Descriptive measures for numerical variables
numerical_measures<- describe(numeric_variables)
sink("C:/yasmeen/Data Science/Project/numeric measures.txt", split = TRUE)
numerical_measures
sink()
variance_covariance_matrix<- var(numeric_variables)
sink("C:/yasmeen/Data Science/Project/Variance covariance matrix.txt", split = TRUE)
variance_covariance_matrix
sink()
correlation_matrix<- cor(numeric_variables)
sink("C:/yasmeen/Data Science/Project/correlation matrix.txt", split = TRUE)
correlation_matrix
sink()
#correlation plot matrix
ggcorrplot(correlation_matrix)

########################(Tables & Plots)#######################################
#Bar plots for categorical train_data
par(mfrow = c(3,3)) #Reset plotting area
barplot(table(train_data$Gender), main = "Gender", ylab = "Count", col = "chocolate")
barplot(table(train_data$Fever), main = "Fever", ylab = "Count", col = "brown")
barplot(table(train_data$Nausea.Vomting), main = "Nausea.Vomting", ylab = "Count", col = "burlywood4")
barplot(table(train_data$Headache), main = "Headache", ylab = "Count", col = "burlywood")
barplot(table(train_data$Diarrhea), main = "Diarrhea", ylab = "Count", col = "wheat")
barplot(table(train_data$Fatigue...generalized.bone.ache), main = " Fatigue...generalized.bone.ache", ylab = "Count", col = "cyan4")
barplot(table(train_data$Jaundice), main = "Jaundice", ylab = "Count", col = "skyblue4")
barplot(table(train_data$Epigastric.pain), main = "Epigastric.pain", ylab = "Count", col = "pink4")
barplot(table(train_data$Baselinehistological.staging), main = "Baselinehistological.staging", ylab = "Count", col = "red4")

#ggpairs for numerical variable
ggpairs(train_data[, c("Age", "BMI", "WBC", "RBC", "HGB",
                       "Plat", "AST.1", "ALT.1", "ALT.48", "RNA.Base", "RNA.EOT")],
        title = "Pair Plot of Numerical Variables",
        lower = list(continuous = wrap("points", alpha = 0.5)),
        upper = list(continuous = wrap("cor", size = 5, color = "darkgray")),
        diag = list(continuous = wrap("densityDiag", alpha = 0.5)))

#Density Plot
ggplot(train_data, aes(x = RNA.Base, fill = Fever, color = Fever)) +
  geom_density(alpha = 0.3) +
  labs(title = "RNA.Base distribution by Fever",
       x = "Fever",
       y = "RNA.Base") +
  theme_minimal()

ggplot(train_data, aes(x = RNA.Base, fill = Headache, color = Headache)) +
  geom_density(alpha = 0.3) +
  labs(title = "RNA.Base distribution by Headache",
       x = "Headache",
       y = "RNA.Base") +
  theme_minimal()

ggplot(train_data, aes(x = RNA.Base, fill = Diarrhea, color = Diarrhea)) +
  geom_density(alpha = 0.3) +
  labs(title = "RNA.Base distribution by Diarrhea",
       x = "Diarrhea",
       y = "RNA.Base") +
  theme_minimal()

ggplot(train_data, aes(x = RNA.Base, fill = Jaundice, color = Jaundice)) +
  geom_density(alpha = 0.3) +
  labs(title = "RNA.Base distribution by Jaundice",
       x = "Jaundice",
       y = "RNA.Base") +
  theme_minimal()
```

```r
## pie chart for categorical variables
f1 <- Freq(train_data$Gender)
p1 <- ggplot(f1, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Gender") +
  scale_fill_manual(values =c("1" = "wheat3", "2" = "ivory2"))
print(p1)
f2 <- Freq(train_data$Fever)
p2 <- ggplot(f2, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Fever") +
  scale_fill_manual(values =  c("1" = "hotpink4", "2" = "hotpink2"))
print(p2)
f3 <- Freq(train_data$Nausea.Vomting)
p3 <- ggplot(f3, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Nausea.Vomting") +
  scale_fill_manual(values = c("1" = "palegreen4", "2" = "palegreen2"))
print(p3)

f4 <- Freq(train_data$Headache)
p4 <- ggplot(f4, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Headache") +
  scale_fill_manual(values = c("1" = "sienna4",  "2" = "sandybrown"))
print(p4)
f6 <- Freq(train_data$Diarrhea)
p6 <- ggplot(f6, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Diarrhea") +
  scale_fill_manual(values = c("1" = "sienna",  "2" = "wheat"))
print(p6)
f7 <- Freq(train_data$Fatigue...generalized.bone.ache)
p7 <- ggplot(f7, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Fatigue...generalized.bone.ache") +
  scale_fill_manual(values = c("1" = "tan4",  "2" = "tan"))
print(p7)
```

```r
f8 <- Freq(train_data$Jaundice)
p8 <- ggplot(f8, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Jaundice") +
  scale_fill_manual(values = c("1" = "skyblue",  "2" = "skyblue4"))
print(p8)
f8 <- Freq(train_data$Epigastric.pain)
p8 <- ggplot(f8, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Epigastric.pain") +
  scale_fill_manual(values = c("1" = "cyan",  "2" = "cyan4"))
print(p8)
f9 <- Freq(train_data$Baselinehistological.staging)
p9 <- ggplot(f9, aes(x = "", y = freq, fill = level)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(perc * 100), "%")), position = position_stack(vjust = 0.5)) +
  labs(x = NULL, y = NULL, fill = NULL) +
  ggtitle("Pie chart of Baselinehistological.staging") +
  scale_fill_manual(values = c("1" = "pink",  "2" = "pink2", "3" = "pink3", "4" = "pink4" ))
print(p9)
grid.arrange(p1, p2, p3, p4, p6, p7, p8, p9,  nrow = 3, ncol = 3)
#Scatter plot
ggplot(train_data, aes(x = RNA.Base, y = RNA.EOT)) +
  geom_point(color= "tan4") +
  labs(title = "Scatter Plot of RNA.Base vs RNA.EOT",
       x = "RNA.Base",
       y = "RNA.EOT")

# Histogram for numeric variable
graph1 <- ggplot(train_data, aes(Age)) +
  geom_histogram(bins=12,fill = "pink", color = "deeppink2") +
  labs(title = "Histogram Age",
       x = "Age")
print(graph1)
graph2 <- ggplot(train_data, aes(BMI)) +
  geom_histogram(bins = 12, fill = "darkslategrey", color = "cyan4") +
  labs(title = "Histogram BMI", x = "BMI")
print(graph2)
graph3 <- ggplot(train_data, aes(WBC)) +
  geom_histogram(bins=12,fill = "salmon4", color = "tan") +
  labs(title = "Histogram WBC",
       x = "WBC")
print(graph3)
graph4 <- ggplot(train_data, aes(RBC)) +
  geom_histogram(bins=12,fill = "cyan4", color = "lightyellow2") +
  labs(title = "Histogram RBC",
       x = "RBC")
print(graph4)
graph5 <- ggplot(train_data, aes(HGB)) +
  geom_histogram(bins=12,fill = "deeppink2", color = "lightyellow2") +
  labs(title = "Histogram HGB",
       x = "HGB")
print(graph5)
```

```
graph6 <- ggplot(train_data, aes(Plat)) +
  geom_histogram(bins=12,fill = "brown4", color = "wheat") +
  labs(title = "Histogram Plat",
       x = "Plat")
print(graph6)
graph7 <- ggplot(train_data, aes(AST.1)) +
  geom_histogram(bins=12,fill = "wheat4", color = "wheat2") +
  labs(title = "Histogram AST.1",
       x = "AST.1")
print(graph7)
graph8 <- ggplot(train_data, aes(ALT.1)) +
  geom_histogram(bins=12,fill = "green4", color = "lightyellow2") +
  labs(title = "Histogram ALT.1",
       x = "ALT.1")
print(graph8)
graph9 <- ggplot(train_data, aes(ALT.48)) +
  geom_histogram(bins=12,fill = "yellow4", color = "lightyellow2") +
  labs(title = "Histogram ALT.48",
       x = "ALT.48")
print(graph9)
graph10 <- ggplot(train_data, aes(RNA.Base)) +
  geom_histogram(bins=12,fill = "deeppink4", color = "lightyellow2") +
  labs(title = "Histogram RNA.Base",
       x = "RNA.Base")
print(graph10)
graph11 <- ggplot(train_data, aes(RNA.EOT)) +
  geom_histogram(bins=12,fill = "black", color = "lightyellow2") +
  labs(title = "Histogram RNA.EOT",
       x = "RNA.EOT")
print(graph11)
grid.arrange(graph1, graph2, graph3, graph4, graph5, graph6, graph7, graph8,
             graph9,graph10,graph11, nrow = 4, ncol = 3)
```

```r
#Frequency tables for gender, fever & Baselinehistological.staging.
f1 <- Freq(train_data$Gender)
print(f1)
f2 <- Freq(train_data$Fever)
print(f2)
f9 <- Freq(train_data$Baselinehistological.staging)
print(f9)

#Create two way table for HCV & Gender
# Create a two-way table for HCV and Gender
table <- table(train_data$HCV, train_data$Gender)
print(table)
chi_square <- chisq.test(table)
print(chi_square)

# Create a three-way contingency table for HCV with Gender & Epigastric.pain
HCV <- train_data$HCV
Epigastric.pain <-  train_data$Epigastric.pain
Gender <- train_data$Gender
three_way_table <- table(HCV, Epigastric.pain, Gender)
print(three_way_table)
three_way_table_margin <- addmargins(three_way_table)
three_way_table_margin
total <- sum(three_way_table)
total
three_way_table_percentage <- (three_way_table_margin/total)*100
three_way_table_percentage <- format(three_way_table_percentage, digits=2, nsmall=2)
three_way_table_percentage
CHM_test <- mantelhaen.test(three_way_table)
CHM_test

train_data$HCV <- ifelse(train_data$HCV == 1, 0, ifelse(train_data$HCV == 2, 1, train_data$HCV))
test_data$HCV <- ifelse(test_data$HCV == 1, 0, ifelse(test_data$HCV == 2, 1, test_data$HCV))

# Logistic Regression Model
logmodel <- glm(HCV ~ Age + Gender + BMI + Fever + Nausea.Vomting + Headache + Diarrhea +
                  Fatigue...generalized.bone.ache + Jaundice + Epigastric.pain + WBC +
                  RBC + HGB + Plat + AST.1 + ALT.1 + ALT.48 + RNA.Base + RNA.EOT +
                  Baselinehistological.staging,
                family = binomial(link = "logit"), data = train_data, control = glm.control(maxit = 100))

#model summary
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/Summary.txt")
summary(logmodel)

stepwise_model <- step(logmodel, direction = "both")
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/stepwise_model.txt")
summary(stepwise_model)

# exponentiation the estimates to get the odds ratios
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/exp.txt")
exp <- exp(stepwise_model$coefficients)
exp

# Check for multicollinearity using VIF
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/vif_stepwise.txt")
vif_stepwise <- vif(stepwise_model)
vif_stepwise
```

```r
# ANOVA for the model
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/ANOVA.txt")
anova(stepwise_model, test = "Chisq")

# Now predict on the test data
predictedlogit_test <- predict(stepwise_model, test_data)

# Predicted probabilities for the test data
predictedprob_test <- plogis(predictedlogit_test)
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/Predicted probabilities.txt")
summary(predictedprob_test)

# Convert predicted probabilities into binary classes with cutoff = 0.5 for test data
predictclass_test <- ifelse(predictedprob_test > 0.5, 1, 0)

# Confusion matrix for test data
CM_test <- table(test_data$HCV, predictclass_test)
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/CM_test.txt")
print(CM_test)

# Confusion Matrix to test data
confusionMatrix <- confusionMatrix(factor(test_data$HCV), factor(predictclass_test))
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/confusionMatrix.txt")
print(confusionMatrix)

# Classification error metrics for test data
err_metric_test <- function(conf_matrix) {
  accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
  sensitivity <- conf_matrix[2, 2] / (conf_matrix[2, 2] + conf_matrix[2, 1])
  specificity <- conf_matrix[1, 1] / (conf_matrix[1, 1] + conf_matrix[1, 2])
  list(accuracy = accuracy, sensitivity = sensitivity, specificity = specificity)}
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/err_metric_test.txt")
err_metric_test(CM_test)

#McFadden's R2 for our model using the pR2 function
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/McFadden.txt")
pscl::pR2(stepwise_model)["McFadden"]

#calculate VIF values for each predictor variable in our model
sink("C:/Users/yi727/OneDrive/Desktop/4th LEVEL/DATA SCIENCE/vif.txt")
car::vif(stepwise_model)

# ROC curve for test data
roc_score_test <- roc(test_data$HCV, predictedprob_test)
plot(roc_score_test, main = "ROC curve", col = "darkred")
auc_value_test <- auc(roc_score_test)
legend("bottomright", legend = paste("AUC =", round(auc_value_test, 2)),
       col = "darkred", lwd = 2)
```

```r
#######################Machine learning Algorithms##############################
#########################1.(Decision tree)#######################################
#Display the model
model_tree <- rpart(train_data$HCV ~ ., data = train_data, method = "class")
model_tree

#Decision Tree Visualization
rpart.plot(
  model_tree,
  box.palette = list("darkolivegreen", "darkolivegreen1", "darksalmon", "darkred"),
  shadow.col = "gray",
  col = "white",
  main = "Decision Tree Visualization")

#the most important variable in splitting the data
importance <- varImp(model_tree)
importance %>%
  arrange(desc(Overall))

#predictions using the model
test_data$HCV <- as.factor(test_data$HCV)
prediction <- predict(model_tree, newdata = test_data, type = "class")
prediction

#confusion matrix
confusionMatrix(test_data$HCV, prediction)

#########################2.(KNN)#################################################
#use KNN for numerical explanatory and categorical response
numeric_train <- train_data[sapply(train_data, is.integer)]
numeric_test <- test_data[sapply(test_data, is.integer)]
train_scaled <- scale(numeric_train)
test_scaled <- scale(numeric_test)

#choosing K
misClassError<-c()
for(i in 1:15){classifier_knn <- knn(train = train_scaled,
                                     test = test_scaled,
                                     cl = train_data$HCV,
                                     k = i);
misClassError[i] <- mean(classifier_knn != test_data$HCV)}
print(misClassError)
which.min(misClassError)

# K = 7
classifier_knn <- knn(train = train_scaled,
                      test = test_scaled,
                      cl = train_data$HCV,
                      k = 7)
misClassError <- mean(classifier_knn != test_data$HCV)
print(paste('Accuracy =', 1-misClassError))
#Confusion Matrix
cm <- table(test_data$HCV, classifier_knn)
print(cm)
```

```r
#########################(Create Function)######################################
#1- function to calculate the difference between RNA.Base & RNA. EOT
My_Function1 <- function (RNA.Base, RNA.EOT){RNA.Base - RNA.EOT}
diff <- My_Function1(train_data$RNA.Base,train_data$RNA.EOT)
print(diff)

#2- function to calculate the average for HCV categories "1
#: Good response to treatment (No virus) & 2: still have the virus"
My_Function2 <- function(data, value_column, category_column) {
  avg_cat1 <- mean(data[[value_column]][data[[category_column]] == 1], na.rm = TRUE)
  avg_cat2 <- mean(data[[value_column]][data[[category_column]] == 2], na.rm = TRUE)
  return(c(avg_cat1, avg_cat2))}
Avg_RNA.EOT <- My_Function2(train_data, "RNA.EOT", "HCV")
print(Avg_RNA.EOT)
```